Section A — Concept Application

1. **Explain the operational differences between static and dynamic routing, and justify why configuring static routes would be the most efficient choice in this scenario.**

Static Routing vs Dynamic Routing

Static Routing : Static routing is a method in which the network administrator manually configures routes in the router's routing table. These routes remain fixed until the administrator changes them.

Dynamic Routing : Dynamic routing automatically learns and updates routes using routing protocols such as RIP, OSPF, and EIGRP. Routers exchange routing information and select suitable paths automatically.

Operational Differences:

Static Routing	**Dynamic Routing**
Routes are configured manually.	Routes are learned automatically.
Requires less CPU and memory.	Requires more CPU and memory.
No routing protocol updates are needed.	Routers exchange routing updates.
Best for small, simple networks.	Best for large, complex networks.
Does not automatically adapt to network changes.	Can adapt to topology changes.

Why is Static Routing the most efficient choice in this scenario?
Static routing is the most efficient choice when the network is small, has a simple and stable topology, and uses only a few routes. It is easy to configure, consumes fewer resources, provides predictable routing, and does not generate routing update traffic. Therefore, it reduces network overhead and simplifies administration.

2. **Compare RIP and OSPF as dynamic routing protocol options for this network, and recommend which one is better suited as the network grows, justifying your choice based on convergence time and scalability.**

Comparison of RIP and OSPF

RIP and OSPF are dynamic routing protocols used to exchange routing information between routers. For a network with five branch routers, both protocols can work, but OSPF is a better choice for future growth.

Feature	RIP	OSPF
Routing type	Distance-vector	Link-state
Metric	Hop count	Cost based mainly on bandwidth
Maximum hops	15 hops	No 15-hop limit
Convergence	Relatively slow	Generally faster
Scalability	Best for small networks	Suitable for medium and large networks
Updates	Periodic updates	Mainly triggered by topology changes, plus periodic refreshes
Configuration	Simple	More complex

1. Convergence Time

Convergence is the time routers take to update their routing information after a network change or link failure.

- RIP: May take longer to detect and respond to network changes because it relies on periodic updates.
- OSPF: Generally converges faster by distributing link-state information and recalculating the best paths.

2. Scalability

RIP is suitable for small networks but has a maximum usable distance of 15 hops. OSPF supports larger networks and can use areas to organize routing information efficiently.

3. Recommendation: OSPF

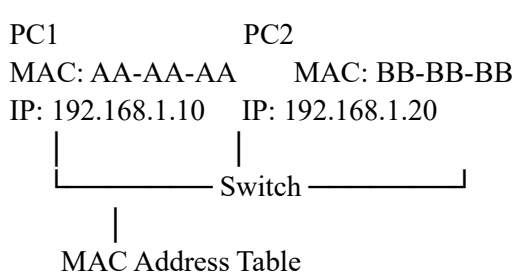
I recommend OSPF for the startup's five branch routers because it offers:

1. Faster convergence after link failures.
2. Better scalability as new branches are added.
3. More flexible path selection using cost.
4. Efficient distribution of routing information.
5. Support for hierarchical network design using areas.

3. **Question: Explain how the switch populates its MAC address table and forwards frames between the two devices, including what happens before the table is populated.**

When **two devices are connected to the same switch** and communicate for the first time, the switch uses **MAC addresses** to learn where each device is located.

Example topology



Initially, the MAC table is empty

When the switch starts, it doesn't know which MAC address belongs to which port.

MAC Address Table

MAC Address	Port
(empty)	

PC1 sends a frame

Suppose PC1 wants to communicate with PC2.

PC1 sends an Ethernet frame:

Source MAC = AA-AA-AA
Destination MAC = BB-BB-BB

The frame enters the switch through **Port 1**.

Switch learns PC1's MAC

The switch looks at the **source MAC address**.

It learns:

"AA-AA-AA came from Port 1."

So it adds:

MAC Address Port

AA-AA-AA Fa0/1

This process is called **MAC address learning**.

But the destination MAC is unknown

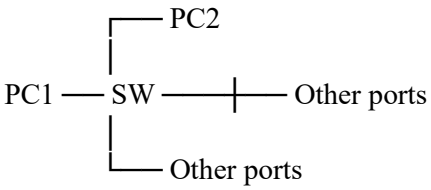
The switch checks:

Destination = BB-BB-BB

But BB-BB-BB isn't in its MAC table yet.

So the switch doesn't know which port PC2 is connected to.

It performs **unknown unicast flooding**.



The switch sends the frame out of the other appropriate ports **except the port where the frame arrived**.

PC2 receives the frame and processes it if the destination MAC matches.

PC2 replies

PC2 sends a reply back to PC1:

Source MAC = BB-BB-BB
Destination MAC = AA-AA-AA

The switch receives this on PC2's port and learns:

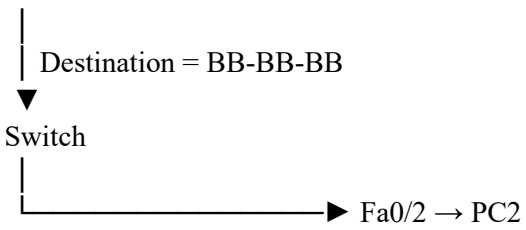
MAC Address Port

AA-AA-AA Fa0/1 BB-BB-BB
Fa0/2

Now the switch knows where both devices are.

Future communication is unicast

If PC1 sends another frame to PC2:



The switch looks up BB-BB-BB in its MAC table and sends the frame **only through Fa0/2**.

It does **not flood the frame** to every port.

A switch learns MAC addresses by examining the source MAC address of incoming Ethernet frames and associating that MAC address with the incoming port. When a destination MAC is not yet in the MAC address table, the switch floods the frame out of all ports except the incoming port. Once the destination device responds, the switch learns its MAC address as well. After both MAC addresses are learned, the switch forwards subsequent frames directly to the correct port instead of flooding them.

4. **Explain the purpose of VLAN segmentation, and describe how an 802.1Q trunk link allows traffic for multiple VLANs to travel between the switch and the router over a single physical connection.**

VLAN Segmentation and 802.1Q Trunking

1. Purpose of VLAN Segmentation

VLAN (Virtual Local Area Network) is a logical method of dividing a physical switch network into multiple separate broadcast domains.

VLAN segmentation allows network administrators to separate devices into different logical networks, even when those devices are connected to the same physical switch.

Why do we use VLANs?

1. Improved security: Separates departments such as HR, Finance, and IT to limit direct communication between them.
2. Reduced broadcast traffic: Broadcast traffic in one VLAN normally remains within that VLAN, reducing unnecessary traffic in other VLANs.
3. Better network management: Devices can be grouped by department, function, or requirement rather than physical location.
4. Improved performance: Reduces the size of each broadcast domain and helps manage network traffic efficiently.
5. Logical separation: Devices in different VLANs require Layer 3 routing to communicate with each other.

Example of VLAN segmentation

Suppose an office has three departments connected to one switch.

Switch 1

One physical switch

VLAN 10

HR

192.168.10.0/24

VLAN 20

Finance

192.168.20.0/24

VLAN 30

IT

192.168.30.0/24

Each VLAN is a separate broadcast domain.

In this example, HR devices belong to VLAN 10, Finance devices belong to VLAN 20, and IT devices belong to VLAN 30. Although all departments use the same physical switch, their Layer 2 traffic is logically separated.

Important: VLANs provide logical separation, but communication between VLANs can still be allowed through a router or Layer 3 switch using appropriate routing and access-control policies.

2. What is an 802.1Q trunk link?

IEEE 802.1Q is a standard that allows Ethernet frames to carry VLAN identification information.

A trunk link carries traffic belonging to multiple VLANs over a single physical connection between network devices, such as a switch and a router.

Without trunking, separate physical connections may be needed for each VLAN. With 802.1Q trunking, one link can carry traffic for VLAN 10, VLAN 20, VLAN 30, and other configured VLANs.

How does a trunk link work?

PCs in different VLANs

VLAN 10 · VLAN 20 · VLAN 30

Managed Switch

Separates traffic by VLAN

802.1Q Trunk

One physical Ethernet link carrying multiple VLANs

Router

Subinterfaces for VLAN 10, 20 and 30

Router-on-a-stick: one physical router interface handles routing for multiple VLANs.

The process works as follows:

1. A PC sends an Ethernet frame to the switch.
2. The switch identifies the VLAN associated with the incoming access port.
3. When the frame leaves through the trunk, the switch normally adds an 802.1Q tag identifying the VLAN.
4. The frame travels across the same physical link as traffic from other VLANs.

5. The router examines the VLAN tag and processes the frame through the corresponding logical subinterface.
6. If communication with another VLAN is required, the router routes the packet and forwards it into the destination VLAN.

The router and switch must have compatible trunk configuration, and the required VLANs must be allowed on the trunk.

What information does an 802.1Q tag contain?

The tag is inserted into the Ethernet frame and includes a 12-bit VLAN ID, which supports VLAN IDs from 0 to 4095, with 1 to 4094 generally available for normal VLAN assignment.

The tag allows the receiving device to identify which VLAN the frame belongs to.

3. Cisco configuration example

Suppose we have:

- VLAN 10: HR
- VLAN 20: Finance
- VLAN 30: IT
- Switch port G0/1: Connected to the router
- Router interface G0/0: Connected to the switch

Switch configuration

```
enable
configure terminal

vlan 10
 name HR
vlan 20
 name FINANCE
vlan 30
 name IT

interface gigabitEthernet 0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
 no shutdown
end
show interfaces trunk
```

This creates the VLANs and configures the switch port as a trunk that allows VLANs 10, 20, and 30. The exact trunk commands may vary by switch model.

Router configuration

```
enable
configure terminal

interface gigabitEthernet 0/0
 no shutdown

interface gigabitEthernet 0/0.10
 encapsulation dot1Q 10
```

```
 ip address 192.168.10.1 255.255.255.0

interface gigabitEthernet 0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0

interface gigabitEthernet 0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
end
show ip interface brief
```

Each router subinterface acts as the default gateway for its VLAN:

VLAN	Department	Default gateway
10	HR	192.168.10.1
20	Finance	192.168.20.1
30	IT	192.168.30.1

This configuration is an example of router-on-a-stick, where one physical router interface handles traffic for multiple VLANs through logical subinterfaces.

5. **Question: Explain why inter-VLAN routing is required here, and justify the specific advantages of a Router-on-a-Stick configuration over adding a second physical router interface.**

Inter-VLAN Routing

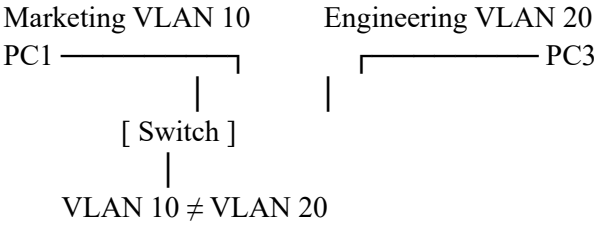
In this scenario, the startup has two VLANs:

- **Marketing VLAN** → e.g. VLAN 10
- **Engineering VLAN** → e.g. VLAN 20
- Both VLANs are connected to the **same switch**
- There is only **one router**

Why is Inter-VLAN Routing required?

By default, **different VLANs are separate Layer-2 broadcast domains**.

For example:



A normal Layer-2 switch can forward frames **within the same VLAN**, but it cannot route traffic from VLAN 10 to VLAN 20.

So if:

PC1 (VLAN 10)
IP: 192.168.10.10
↓
PC3 (VLAN 20)
IP: 192.168.20.10

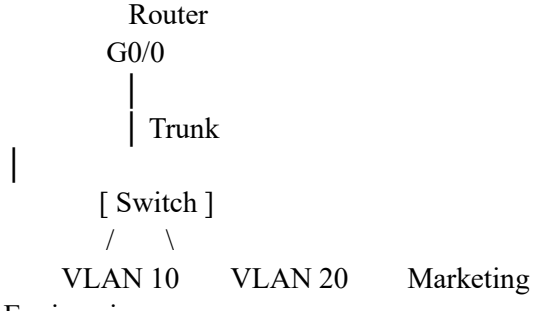
PC1 needs a **Layer-3 device** to move the packet between the two different networks.

That is the job of **inter-VLAN routing**.

Why Router-on-a-Stick?

Instead of adding another physical router interface, we can use **one physical router interface divided into multiple logical subinterfaces**.

Example:



On the router:

G0/0.10 → VLAN 10 → 192.168.10.1
G0/0.20 → VLAN 20 → 192.168.20.1

The switch sends VLAN-tagged traffic through the single trunk link, and the router's subinterfaces route traffic between the VLANs.

Main advantages

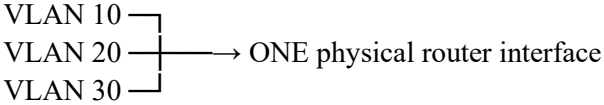
1. Saves hardware

You don't need a separate physical router interface for every VLAN.

Traditional:
VLAN 10 → Physical Interface
VLAN 20 → Physical Interface

VLAN 30 → Physical Interface

With Router-on-a-Stick:



2. Lower cost

For a small startup, using one existing router avoids buying additional router hardware or interface modules.

3. Easy to expand

If the company adds another VLAN, you can create another subinterface rather than installing another physical interface.

Example:

G0/0.10 → Marketing
G0/0.20 → Engineering
G0/0.30 → HR
G0/0.40 → Finance

4. Centralized configuration

All inter-VLAN routing can be configured on the same router.

Router-on-a-Stick vs Multiple Physical Interfaces

Router-on-a-Stick	**Multiple physical interfaces**
One physical interface	One physical interface per VLAN
Uses subinterfaces	Uses physical interfaces
Lower hardware cost	More hardware required
Easy to add VLANs	Limited by available interfaces
Requires trunk link	Separate links/interfaces
Suitable for small networks	Can provide better performance at higher traffic

One important disadvantage

Router-on-a-Stick has **one physical link carrying traffic for multiple VLANs**. Therefore, that link can become a bottleneck if the network has heavy inter-VLAN traffic.

For this **small startup with a single router**, however, Router-on-a-Stick provides a practical way to enable Marketing ↔ Engineering communication without adding another physical router interface.

“Inter-VLAN routing is required because VLANs are separate Layer-2 broadcast domains and a Layer-2 switch cannot route traffic between them. Router-on-a-Stick allows one physical router interface to provide routing for multiple VLANs using logical subinterfaces over a trunk link, reducing hardware cost and making the network easier to scale.”